SUTD404: Denoising Language Models for Room-to-Address Lookup

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- SUTD has over 200 rooms across 5 buildings.
- Rooms are often communicated by name.
- Room-to-address lookup can be a hassle.

- Resources for this task remain limited.
- Physical signs often require one to walk around.
- Digital resources like the SUTD Assistive Maps (SAM) Telegram bot often fail on noisy input (e.g. strings with typos).

- We introduce SUTD404 to address this gap.
- Visit **github.com/gregorylimeurhen/sutd404** for our code repository.
- Visit **sutd404.vercel.app** for our web application.

Definition (Keyboard)

$$K = \begin{bmatrix} ‘ & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 0 & - & = \\ q & w & e & r & t & y & u & i & o & p & [&] & \backslash \\ & a & s & d & f & g & h & j & k & l & ; & ’ & \\ & & z & x & c & v & b & n & m & , & . & / & \\ & & & & & & \sqcup & \sqcup & \sqcup & \sqcup & \sqcup & & \end{bmatrix}$$

- We define a 60% ANSI QWERTY en-US keyboard layout.

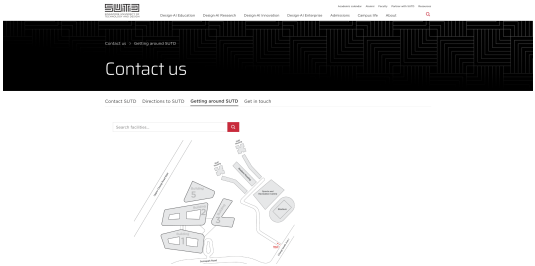

Figure: Official SUTD contact pages.

- We obtain a room set R by politely scraping official SUTD contact pages.
- We queried the facility search function by enumerating buildings and floors.

Let V be a vocabulary, and $R \subseteq V^*$ be a room set, and $\prec$ be a total order on R, and $N \in \mathbb{N}_{\geq 1}$ be a result count, and $\prec_N$ be a total order on $R^N_{\prec}$, and $A \subseteq V^*$ be an address set, and $B \subseteq V$ be a boundary set, and $C \in \mathbb{N}$ be a corruption count, and $\alpha \in \mathbb{R}_{[0,1]}$ be a transposition rate, and $\beta \in \mathbb{R}_{[0,1]}$ be a substitution rate, and $\boldsymbol{\pi} = (\pi_{\mathrm{train}}, \pi_{\mathrm{val}}, \pi_{\mathrm{test}}) \in \Delta^2_C$ be a training-validation-testing split, and $I : V \rightharpoonup \mathbb{Z}^2$ be a keyboard-position map.

Algorithm 1 CORRUPT($\boldsymbol{x}$)

Require: $\boldsymbol{x} \in V^*$

$(T_L, T_M, T_R) \leftarrow (V^* B \cup \{()\}, (V \setminus B)^+, BV^* \cup \{()\})$

$T \leftarrow \{\boldsymbol{uw}b\boldsymbol{vz} \mid \boldsymbol{u} \in T_L,\ \boldsymbol{v}, \boldsymbol{w} \in T_M,\ \boldsymbol{z} \in T_R,\ b \in B,\ \boldsymbol{x} = \boldsymbol{uv}b\boldsymbol{wz}\} \setminus \{\boldsymbol{x}\}$

if $T \neq \varnothing$ **then** $\boldsymbol{x} \sim \alpha\,\mathcal{U}(T) + (1-\alpha)\,\delta(\boldsymbol{x})$

for $i \in [1..|\boldsymbol{x}|]$ **do**

  $S \leftarrow \{k \in K \setminus \{x_i\} \mid I(k) - I(x_i) \in [-1..1] \times [-2..2]\}$

  if $S \neq \varnothing$ **then** $x_i \sim \beta\,\mathcal{U}(S) + (1-\beta)\,\delta(x_i)$

return $\boldsymbol{x}$

- We generated corrupted strings and split corrupted-uncorrupted room pairs into training, validation, and testing datasets:
  - Our corruption algorithm transposes spans and substituting characters.
  - We set $C = 50$, and $\alpha = 0.5$, and $\beta = 0.05$, and $\boldsymbol{\pi} = (0.8, 0.1, 0.1)$ for tractable and plausible-looking typos.
  - We stratify the split by room to make the datasets comparable across rooms.

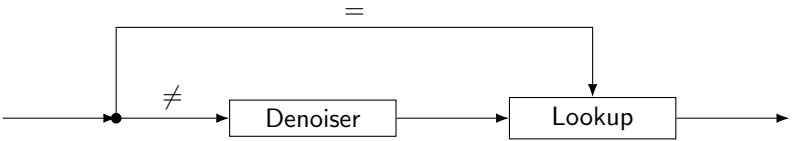

Figure: Inference algorithm.

Definition (Denoiser)

Let $f : V^* \times R \to \mathbb{R}$ be a scorer.

$$\mathscr{D}_f : V^* \to R$$
$$\mathscr{D}_f(x) = \min_{\prec} \arg\max_{r \in R} f(x, r)$$

- Our denoiser is a Nanochat-parameterised, character-level transformer of depth 2 with decoding constrained to the character-level trie induced by R (Karpathy, 2025).
  - Character-level tokenisation: As room strings/typos are often short/character-level.
  - Trie-constrained decoding: To efficiently avoid room hallucination.
- We train our denoiser using a standard cross-entropy loss and AdamW optimiser.[1]

[1] A more complete list of hyperparameters can be found in our code repository.

Definition (Lookup)

$$\mathscr{L} : R \to A$$

- Our lookup is a standard room-address table.

Denoiser	**Accuracy** (%)	**Mean Latency** (ms)
LCP	29.2	**0.04**
DL	71.9	16.7
SJ	89.3	20.8
Ours	**97.3**	8.19

Table: Denoiser evaluation results summary.

- We evaluated our denoiser against substring Jaccard similarity (SJ), negative Damerau–Levenshtein distance (DL), and longest common prefix length (LCP) denoisers.
- We measured accuracy and mean latency of each denoiser on test set.
- We fixed a seed and hardware for fairness and reproducibility.

Algorithm 2 $\textsc{Predict}(x)$

Require: $x \in V^*$

$x \leftarrow \textsc{Alias}(\textsc{AlphanumericSplit}(\textsc{Lowercase}(\textsc{Depunctuate}(\textsc{Strip}(x)))))$

if $x \in R$ **then return** $((x, \mathscr{L}(x)))$

$\boldsymbol{f} \leftarrow (\mathrm{LCP}, \mathrm{SJ}, \mathrm{DL}, \textsc{Ours})$

for $r \in R$ **do**

  for $f \in \boldsymbol{f}$ **do** $p(r, f) \leftarrow |\{r' \in R \mid f(x, r') > f(x, r)\}| + 1$

for $r \in R$ **do** $s(r) \leftarrow \sum_{i=1}^{|\boldsymbol{f}|} (1/|R|^{i-1})(|R| - p(r, f_i))$

$\boldsymbol{o} \leftarrow \min_{\prec_N} \arg\max_{\boldsymbol{r} \in R_{\prec}^N} \sum_{i=1}^{N} s(r_i)$

return $((o, \mathscr{L}(o)) \mid o \in \boldsymbol{o})$

- Our system makes a prediction by:
  1. Normalising by stripping whitespace and punctuation, lowercasing, separating letters from numbers, and replacing words with their aliases.
     - Example: `␣LT,2␣-> LT,2 -> LT2 -> lt2 -> lt 2 -> lecture theatre 2`.
  2. Denoising and looking up with harmonically weighted Borda ranking.

- To further improve the user experience by making the application more efficient and responsive, we implemented:
  - **Incremental prediction.** We predict during typing rather than waiting for final submission.
  - **Prefix caching.** We KV-cache previously computed prefix states and only process newly appended characters.
  - **Debounced querying.** We introduce a short debounce to avoid redundant intermediate inference on rapid keystrokes.
  - **Redundant-query suppression.** We skip empty, already displayed, and exact-match queries when full recomputation is unnecessary.
  - **Bounded result caching.** We memoise recent solved queries while capping cache growth to prevent performance degradation over time.

- We evaluated our system against SAM.
- We measured accuracy on a held-out set of 10 inputs sampled from our test set.
- We fixed the same seed and hardware as before for fairness and reproducibility.

Input	`design bdhavioural laboratory 1`
Gold	`behavioural design laboratory 1`
Prediction	–
Input	`capstone rolm 8`
Gold	`capstone room 8`
Prediction	`capstone room 8`, `capstone room 7`, `capstone room 6`, `capstone room 5`
...	...
Input	`cohference capsrone ro9m 2`
Gold	`capstone conference room 2`
Prediction	`capstone room 9`

Table: SAM evaluation results.

System	Accuracy (%)
SAM	40
Ours	**100**

Table: System evaluation results summary.

- SAM occasionally recovered the target, but degraded as noise became more severe.
- Our system outperformed SAM, achieving complete recovery.
- We believe these results demonstrate strong promise and provide practical foundations for reliable campus assistance.

Notation

Let $n \in \mathbb{N}_{\geq 1}$.

$$R^n_{\prec} = \{\boldsymbol{r} \in R^n \mid r_1 \prec \cdots \prec r_n\}$$

Notation

Let $c \in \mathbb{N}$.

$$\Delta^2_c = \left\{ \boldsymbol{p} \in \mathbb{R}^3_{[0,1]} \;\middle|\; \sum_{i=1}^{3} p_i = 1,\ c\boldsymbol{p} \in \mathbb{N}^3 \right\}$$

Andrej Karpathy. nanochat: The best chatgpt that $100 can buy. https://github.com/karpathy/nanochat, 2025.